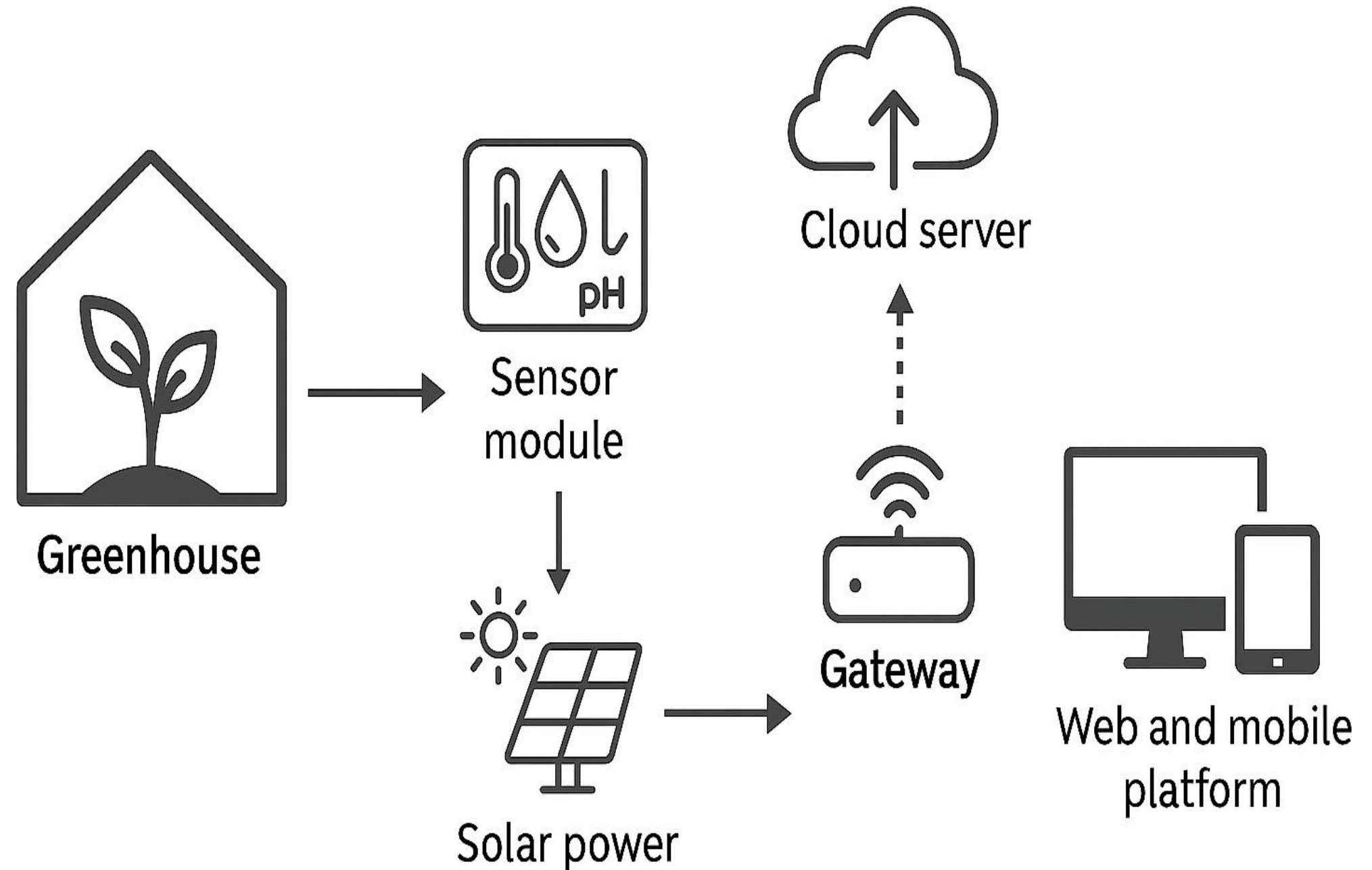


Solving an Engineering Problem

A simple but very useful solution to a problem that affects flower growers in Colombia

System Diagram



Introduction

I want to share a simple but very useful solution to a problem that affects flower growers in Colombia.

Many greenhouses do not have an easy way to know what is happening inside in real time. My goal is to show how technology can help monitor conditions such as temperature, humidity, and soil quality, so farmers can make better decisions and use their resources more efficiently.

This project connects engineering with agriculture by combining sensors, communication, and data to make farming smarter and more sustainable.



Problem Description

No Monitoring Systems

Many flower farms in Colombia work without systems that monitor their internal environment. Farmers do not have continuous information about humidity, temperature, sunlight, or soil pH.

Cannot React in Time

Because of this, they cannot react in time when conditions change. For example, they might not open the ventilation, adjust irrigation, or turn on heating when needed.

Negative Consequences

This can cause plant stress, waste of water and energy, and lower production quality.



Proposed Solution

The proposed solution is an **IoT-based system** made up of several sensor devices placed inside the greenhouse. Each sensor measures temperature, humidity, sunlight, soil moisture, and pH.



Solar Power

These devices use solar power to work independently and send data through LoRa communication, a low-power network that is ideal for rural areas.



Cloud Storage

The data goes from the sensors to a gateway, and then to a cloud server, where it is stored and processed.



Real-Time Access

Farmers and managers can see all the information through a web and mobile platform in real time, with simple charts and reports.

The system also sends alerts through WhatsApp or email when something unusual happens, so users can act right away.



Implementation Plan

The project will follow several stages, based on both the software development cycle and the hardware design process. The estimated total project duration is 8-12 months.

01

Planning Phase (2-3 weeks)

This phase defines what the system will do, which environmental parameters will be measured, and what kind of users will use it (operators, technicians, or managers). We also create a general schedule and estimate the human, technical, and financial resources required.

02

Design Phase (4-6 weeks)

The design process is divided into hardware and software parts:

For hardware, we select sensors (for temperature, humidity, light, pH, etc.), microcontrollers, and LoRa communication modules. We also design the solar power supply and protective casing for outdoor use.

For software, we define the communication architecture, database structure, and user interfaces. We plan how the data will move from the greenhouse sensors to the cloud and create initial mockups of the web and mobile app.

03

Development Phase (8-12 weeks)

Hardware: Build the sensor prototypes, connect the components, program the microcontrollers, and test power use and connectivity.

Software: Develop the cloud platform to store data, create web services to link the gateway with the cloud, and design the user-friendly web and mobile apps. Visualization tools and automatic notifications are also included.

04

Integration and Testing Phase (6-8 weeks)

All system components are tested together: sensors, LoRa network, gateway, internet connection, and the cloud platform. We check that the data is accurate under different environmental conditions (day/night, humidity, temperature variations) and test system alerts to confirm reliability.

Anticipated Challenges

Implementing an IoT-based system in a greenhouse environment can present several obstacles. Addressing these proactively will be key to success.

- Weather Conditions:** Outdoor equipment must withstand varying weather, including extreme temperatures, humidity, and precipitation. This requires robust casing and reliable components.
- Internet Connectivity:** Rural areas often have unstable or limited internet access, which can hinder data transmission to the cloud and real-time monitoring capabilities.
- User Adoption & Training:** Farmers and staff may require extensive training to effectively use the new platform and integrate it into their daily operations.
- Sensor Calibration & Maintenance:** Regular calibration and maintenance of sensors are crucial to ensure data accuracy over time, requiring scheduled checks and potential replacements.
- Budget Constraints:** Small farms may face financial limitations, making the initial investment and ongoing operational costs a significant concern.



Implementation Plan (continued)

1

Field Implementation Phase (4-6 months)

Installation and configuration of the system in pilot greenhouses, including sensors, network, and gateway. Continuous performance monitoring, real data analysis, and parameter adjustments. User feedback is collected, and regular maintenance is performed to ensure stable and efficient operation.

2

Training and Support Phase (2-3 weeks)

Users such as farmers and technicians are trained on how to use the platform, understand the data, and respond to alerts. Basic documentation is provided for system maintenance and troubleshooting.

3

Maintenance and Continuous Improvement Phase (Ongoing)

After installation, the system is kept up to date. Sensors are recalibrated regularly, software updates are released, and new features are added based on user feedback.

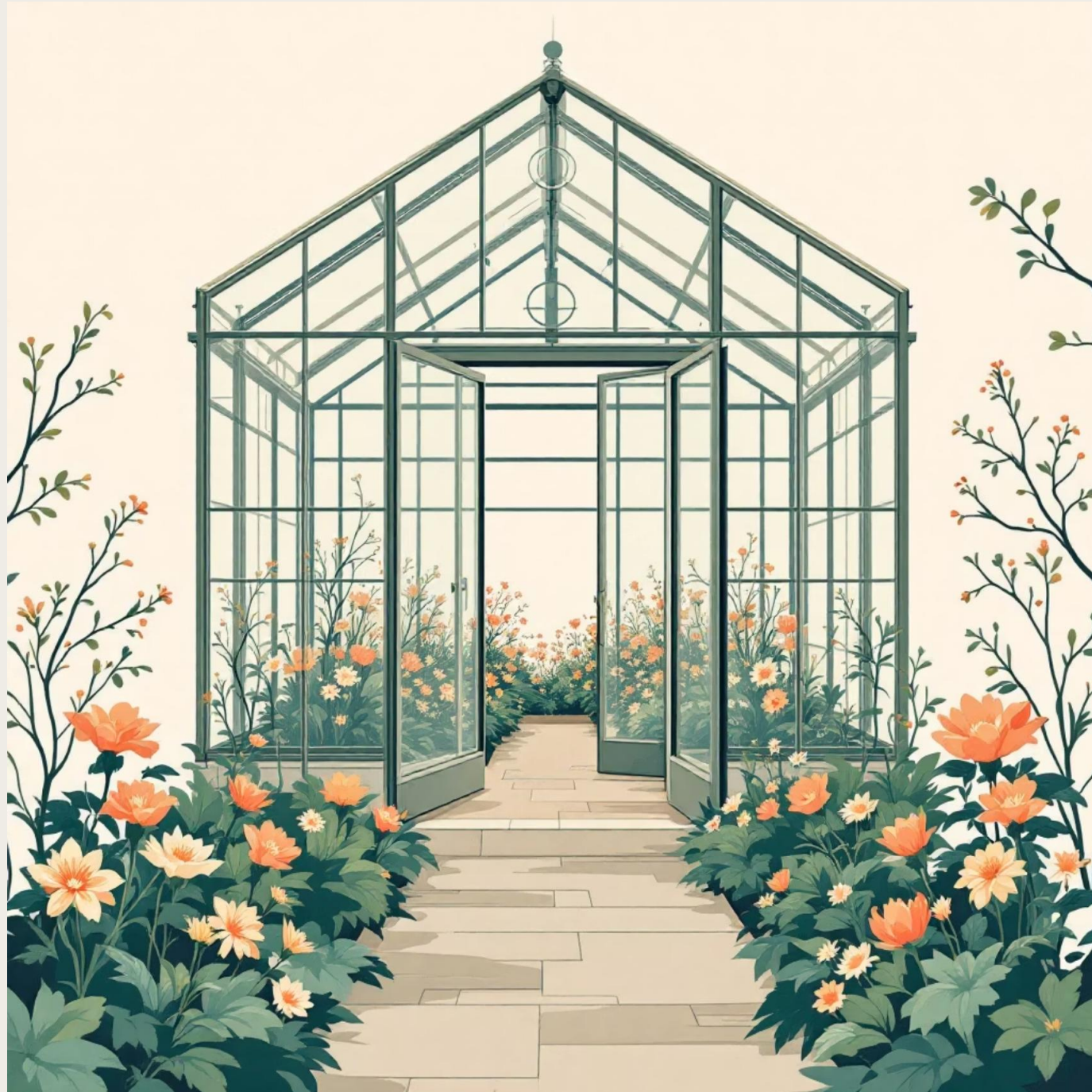
Additional Challenges for Later Phases

As the project progresses into later stages, new challenges may emerge that require careful consideration and proactive solutions.

- **Resistance to Change:** Overcoming reluctance from users accustomed to traditional farming methods to adopt new technological solutions.
- **Data Interpretation Difficulties:** Ensuring non-technical users can effectively understand and utilize the data provided by the system to make informed decisions.
- **Long-term Hardware Durability:** Maintaining the functionality and accuracy of hardware components in harsh greenhouse conditions over extended periods.
- **Scaling Challenges:** Effectively expanding the system to accommodate additional greenhouses or multiple locations without compromising performance.
- **Ongoing Technical Support:** Providing continuous and responsive technical assistance to address user issues and system anomalies.



Conclusion



Throughout this presentation, we've explored the challenges of greenhouse management and introduced our innovative IoT solution. By leveraging **smart sensors, solar power, and cloud connectivity**, we provide a comprehensive system for monitoring greenhouse conditions effectively.

This approach offers clear benefits: **cost-effectiveness and real-time data access** empower farmers to make informed decisions. This leads to improved resource efficiency in managing water, temperature, and light, directly contributing to **reduced waste, increased productivity, and the promotion of sustainable farming practices**.

Our structured implementation plan ensures a smooth transition and long-term success. We believe this smart and affordable technology is key to a more productive and environmentally friendly future in agriculture.

Thank you for your attention. We are excited about the potential of this project and welcome further discussion on how we can implement this solution for you.





Q&A

Questions and Answers

I will now be happy to answer any questions you may have about the system, its design, or how it can be applied in real greenhouses. Feel free to ask about the technology, the implementation process, or the benefits it can bring to the agricultural sector.

